Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Publication

- *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
ML / Deep Learning	PyTorch, HuggingFace Transformers, vLLM, JAX, NumPy, scikit-learn, language modeling, mechanistic interpretability, activation patching, autodiff, normalizing flows, diffusion, transfer learning
Evaluation & NLP	evaluation methodology, anti-Goodharting guardrails, holdout protection, bootstrap CIs / McNemar's test, RAG, RRF, cross-encoder reranking, MMR, sentence-transformers, Ollama, llama-index, spaCy
Backend & Systems	FastAPI, Node.js, REST APIs, async services, Linux profiling, Apache Spark / PySpark, MongoDB, PostgreSQL, Redis, SQLite
Cloud & DevOps	Azure, AWS, GCP, Docker, Git, GitHub Actions, CI/CD

Research & Projects

Refusal Decay - Mechanistic Interpretability of LLM Refusal — github.com/Deva-1903/refusal-decay
Python, PyTorch, HuggingFace Transformers, Llama-3.1-8B-Instruct | UMass COMPSCI 602 Research

- Ran a mechanistic-interpretability study (graded full marks) of how prefilling attacks weaken refusal in **Llama-3.1-8B-Instruct**, extracting the residual-stream refusal direction via difference-in-means on a held-out prompt set; behavioral refusal dropped from **0.92 to 0.32** at prefill length k=3.
- Designed two causal interventions (cross-condition activation patching and additive direction injection via **PyTorch forward hooks**) with multi-seed random/orthogonal controls and a benign positive control; reported a clean null (**0/300** prompts recovered at late layers) with bootstrap 95% CIs and McNemar's test, isolating the late-layer signal as a readout rather than the causal lever.

Generative Models & Deep Learning — github.com/Deva-1903/UMASS_CICS_689
JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented a **matrix-based reverse-mode automatic differentiation engine** from scratch in NumPy (computation graphs, topological-sort backpropagation, operators including matmul, linear solve, and log-determinant), validated against JAX to machine precision; derived convergence guarantees for gradient descent and SGD on PSD/PD objectives.
- Implemented generative models from first principles in JAX, a **normalizing flow with coupling layers** and a **DDPM diffusion model**, deriving the ELBO and closed-form Gaussian-KL objectives via variational inference; trained and benchmarked **5 architectures** on CIFAR-10 across 3 optimizers.

AutoEval - Self-Improving Evaluation Agent — github.com/Deva-1903/AutoEval
Python, IR Evaluation, Git Automation

- Built a **self-improving evaluation agent** that proposes new queries and labels, scores them against multiple reference systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective.
- Added **anti-Goodharting guardrails** (schema checks, duplicate rejection, read-only holdout protection) and committed accepted updates back to Git with report cards for an auditable, reproducible self-improvement loop.

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; replaced spreadsheet workflows and accelerated iteration on downstream NLP/clinical models.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability and remediated **20+ VAPT findings**.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.